COMPASS Coastal Observations, Mechanisms, and Predictions Across Systems and Scales	COMPASS Field Sampling Protocol Chesapeake Bay Region	
Date Created: 21 Sept 2022 Date Updated: 17 Jan 2023	Creator Name(s): A. Stearns Editor Name(s): A. Stearns	Version: 2

Vegetation Plot Installation and Sampling

Objective:

To establish replicate vegetation plots that will allow COMPASS to monitor a representative sample of plants and herbaceous ground cover over time.

Contents:

I. Experimental Design:

Each Chesapeake Bay Synoptic Site will have five (5) vegetation plots installed in each zone (Wetland, Upland and Transition). These plots will be monitored annually, during peak growth. In the Chesapeake Bay region, peak growth occurs in the transition between the July and August months.

Vegetation plots will be either 1m^2 or 2m^2, depending on the zone, and be sectioned with a 10cm PVC grid system.

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Vegetation Plot Installation and Sampling

Vegetation plots will be annually surveyed using the Braun Blanquet method to determine species density, abundance, and frequency.

Annual vegetation harvests will occur during peak growth. Vegetation from a randomly chosen 10cm grid square from each vegetation plot will be collected, dried, and weighed for biomass. Any harvested grid square will not be harvested again for five (5) years.

Vegetation harvests will be sorted by species, stems counted, dried, weighed, and grinded for further nutrient analysis.

II. Personal Protective Equipment: Close-toed shoes and long pants are required at all times when working at COMPASS sites. Rubber boots are strongly recommended in the Wetland and Transition zones. Chest waders are available upon request. Eye protection is available and *is required* in the marsh zones.

III. Installation

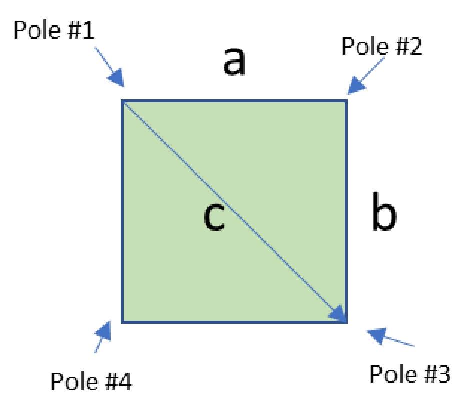
A. Materials:

- (3) metric tapes (at least 3M in length)
- Driveway stakes or plastic stemmed flags - 4/plot, 20/zone, or 60/site
- String
- Scissors
- PVC grids
- Vegetation Plot field notebook
- Pencil
- Sharpie

B. Procedure:

1. Designate location of the vegetation plot
2. Set corner pole #1

3. Pull tape in the direction you want to the distance you want the width. *Installing plots to the cardinal direction is always preferred, if possible.*
4. Set corner pole #2
5. From corner pole #2, pull tape to the distance desired for the length, approximating a right angle, in the proposed direction of pole #3
6. Use the Pythagorean Theorem to find the hypotenuse
   Pull tape from pole #1 toward the proposed position of pole #3
7. Place pole #3 where the length of the side intercepts the length of the hypotenuse
8. Pull tapes from pole #'s 1 and 3 to find the placement of the last corner, pole #4
9. Install the PVC grid



$$a^2 + b^2 = c^2$$
$$c = \sqrt{(a^2 + b^2)}$$

IV. Sampling

A. Materials

- Vegetation Plot field notebook
- Pencil

- Sharpie
- Varying sizes of paper bags (pre-tared bags are recommended)
- Clippers or cutting blade
- Braun-Blanquet score card

B. Procedure

1. Using the Braun-Blanquet score card, conduct a vegetation survey. Document species scores in the field notebook.
2. Using a random number application, select the 10cm grid square to harvest. If this happens to be a grid that has been harvested in the last five (5) years, select another grid square.
3. Label a paper bag, appropriately sized for the vegetation present.
    - Data
    - Site
    - Zone
    - Plot
    - Grid #
    - Species
4. Clip the vegetation as close to the ground as possible.
5. Sort live stems by species. Dead stems can be combined. *This step can be done either in the field, or at the lab (if within a reasonable amount of time)*
6. Count stems by species
7. Place vegetation in the correctly labeled bag
8. Put samples in the grieve oven at 60°C for at least 72 hours

V. Processing

A. Materials:

- Samples
- Balance
- Weights - 1g and 50g standards
- Weigh boats
- Data collection sheet

- Pencil
- Lab gloves

B. Procedure:

1. All samples should be dried in the grieve oven at 60°C for at least 72 hours
2. Pull samples out of the oven and let them cool for about 20 minutes
3. Turn on the balance 10 minutes prior to starting
4. Make sure the balance is level
5. Weigh and record the mass of 1g and 50g standards. Be sure to weigh the standards each new day when using the balance
6. Weigh the sample. If the bag has been pre-tared, you can weigh the sample in the bag, documenting the mass of the bag on the data sheet. If not, remove the entire sample from the bag and use a weigh boat to obtain the samples' mass.
7. Record the mass on the data collection sheet or into the spreadsheet

VI. CB Site Installation Logistics

A. Goodwin Islands

- Installed September 2022
- Wetland: 1m^2 vegetation plots South of the Wetland logger and MET station. Installed along a 30m transect East to west approximately 3m apart from each other. Braun-Blanquet scores obtained. Harvested the NE 10cm grid square for processing.
- Transition: Braun-Blanquet score was obtained from the macrophyte collars. No harvest was done. No additional plots installed for 2022.
- Upland: No plots were installed for 2022; Upland installed 20 July 23. Used the 20m western edge of the Tree Plot as our x-axis and used a random number app to obtain x-y coordinates to indicate the SW corner pole of each plot. Five (5) 2m^2 plots were installed
    - V1 (2,16)
    - V2 (3,2)
    - V3 (6,7)
    - V4 (15,12)
    - V5 (18,19)

B. Moneystump Swamp

- Installed September 2022
- Wetland: 1m^2 vegetation plots West of the Wetland logger and MET station. Installed along a 30m transect East to west approximately 3m apart from each other. Braun-Blanquet scores obtained. Harvested the NE corner 10cm grid square for processing.
- Transition: 2m^2 vegetation plots were installed. Plot locations were selected using random numbers to obtain an x,y pace. Braun-Blanquet scores observed. Harvested the SW corner 10cm grid square for processing.
- Upland: 2m^2 vegetation plots were installed. Plot locations were selected using random numbers to obtain an x,y pace. Braun-Blanquet scores observed. Harvested the ??SW corner 10cm grid square for processing.

C. GCReW

- Installed October 2022
- Wetland: Vegetation data will be compiled from GCReW's annual plant Census
- Transition: 2m^2 vegetation plots were installed. Plot locations were selected using random numbers to select (x,y) coordinates from an installed N-S baseline. Braun-Blanquet scores observed. Harvested the SW 10cm grid square for processing.
- Upland: Vegetation data will be compiled from TEMPEST'S bi-annual herbaceous ground cover Census

D. Sweethall Marsh